AI ASSITANT CODING

ASSIGNMENT – 12.5

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Batch – 35

Task Description #1 (Sorting – Merge Sort Implementation)

• Task: Use AI to generate a Python program that implements the Merge Sort algorithm.

• Instructions:

o Prompt AI to create a function merge_sort(arr) that sorts a list in ascending order.

o Ask AI to include time complexity and space complexity in the function docstring.

o Verify the generated code with test cases.

```python
def merge_sort(arr):
    """
    Sorts a list in ascending order using the Merge Sort algorithm.

    Time Complexity:
    - Best Case: O(n log n)
    - Average Case: O(n log n)
    - Worst Case: O(n log n)

    Space Complexity:
    - O(n) auxiliary space for temporary arrays used during merging.
    - Recursive stack space: O(log n)

    Parameters:
    arr (list): List of elements to be sorted.

    Returns:
    list: New sorted list in ascending order.
    """

    if len(arr) <= 1:
        return arr

    # Find middle index
    mid = len(arr) // 2

    # Divide the array into two halves
    left = merge_sort(arr[:mid])
    right = merge_sort(arr[mid:])

    # Merge the sorted halves
    return merge(left, right)


def merge(left, right):
    result = []
    i = j = 0

    # Compare elements from left and right arrays
    while i < len(left) and j < len(right):
        if left[i] <= right[j]:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
            j += 1

    # Add remaining elements
    result.extend(left[i:])
    result.extend(right[j:])

    return result
```

```
# -------- Test Cases --------

def test_merge_sort():
    print("Running Test Cases...\n")

    test_cases = [
        [38, 27, 43, 3, 9, 82, 10],
        [5, 2, 9, 1, 5, 6],
        [],
        [1],
        [10, 9, 8, 7, 6],
        [3, 3, 3, 3],
        [-5, -10, 0, 5, 2]
    ]

    for i, case in enumerate(test_cases, 1):
        sorted_arr = merge_sort(case)
        print(f"Test Case {i}:")
        print("Original:", case)
        print("Sorted:  ", sorted_arr)
        print("-" * 30)


# Run tests
test_merge_sort()
```

```
Running Test Cases...

Test Case 1:
Original: [38, 27, 43, 3, 9, 82, 10]
Sorted:   [3, 9, 10, 27, 38, 43, 82]
------------------------------
Test Case 2:
Original: [5, 2, 9, 1, 5, 6]
Sorted:   [1, 2, 5, 5, 6, 9]
------------------------------
Test Case 3:
Original: []
Sorted:   []
------------------------------
Test Case 4:
Original: [1]
Sorted:   [1]
------------------------------
Test Case 5:
Original: [10, 9, 8, 7, 6]
Sorted:   [6, 7, 8, 9, 10]
------------------------------
Test Case 6:
Original: [3, 3, 3, 3]
Sorted:   [3, 3, 3, 3]
------------------------------
Test Case 7:
Original: [-5, -10, 0, 5, 2]
Sorted:   [-10, -5, 0, 2, 5]
------------------------------
```

Explaination : The merge_sort() function divides the list into smaller halves recursively until each part has one element, then the merge() function combines these parts by comparing elements to produce a sorted list in ascending order.

Complexity: Time Complexity = O(n log n), Space Complexity = O(n).

Task Description #2 (Searching – Binary Search with AI Optimization)

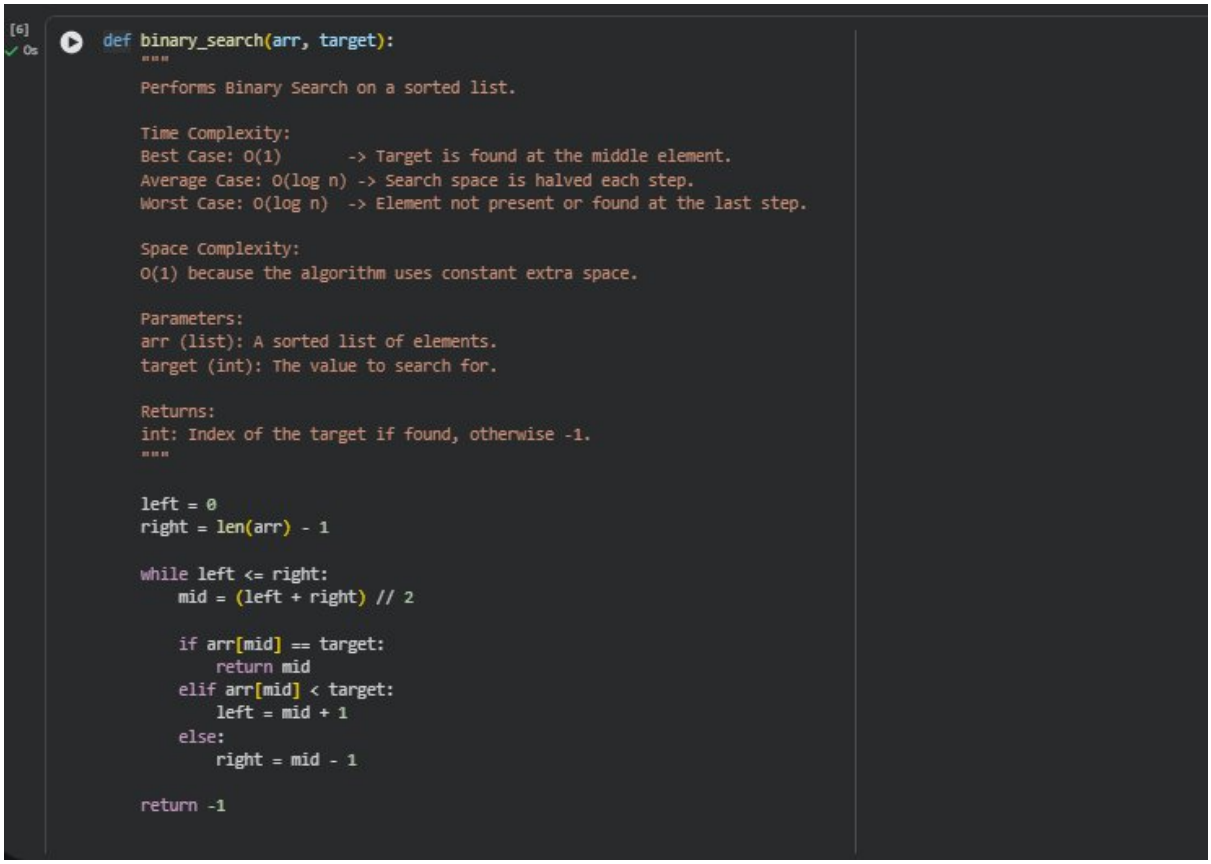
• Task: Use AI to create a binary search function that finds a target element in a sorted list.

• Instructions:

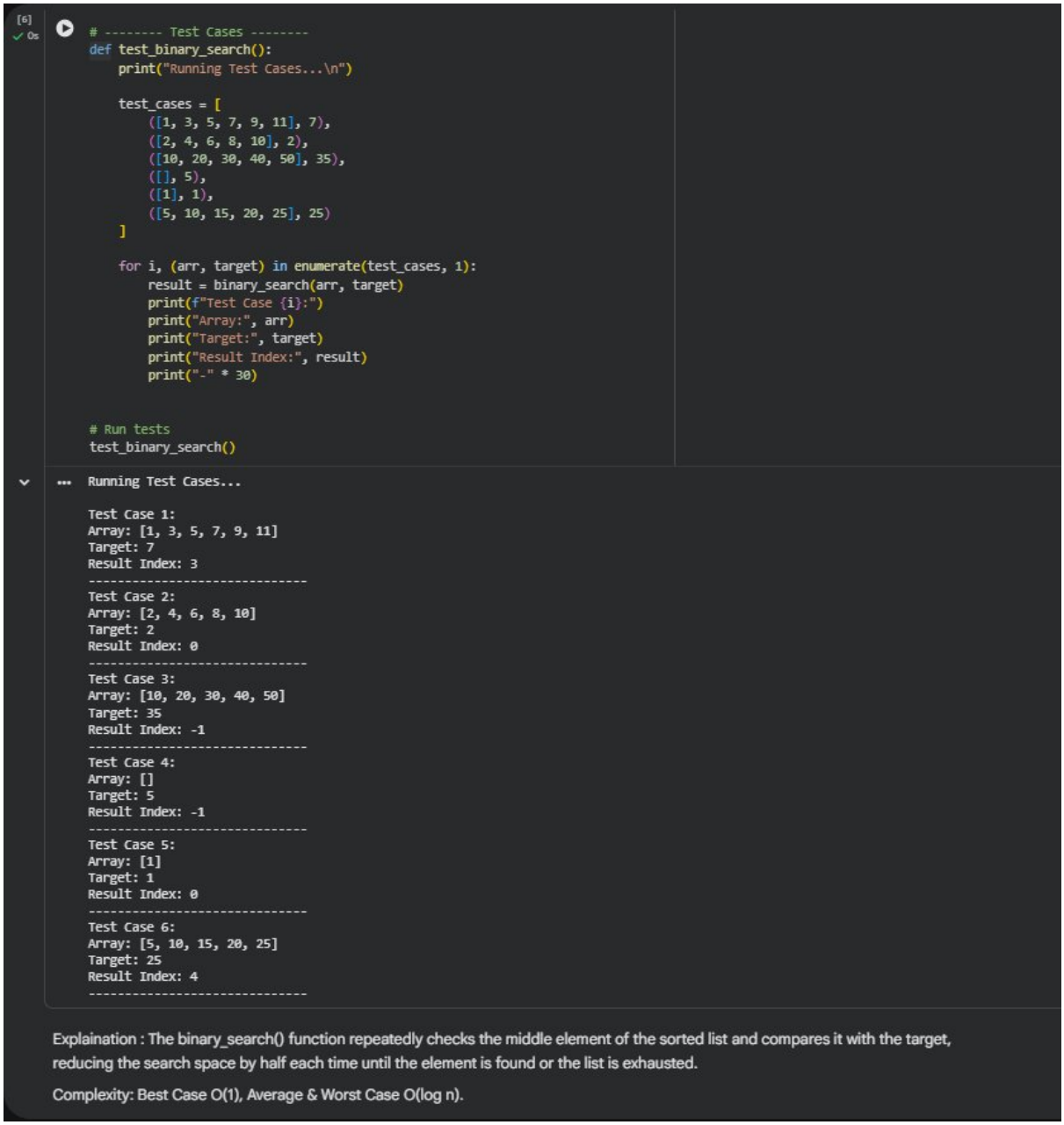
o Prompt AI to create a function binary_search(arr, target) returning the index of the target or -1 if not found.

o Include docstrings explaining best, average, and worst-case complexities.

o Test with various inputs.



```
[6]
✓ 0s
def binary_search(arr, target):
    """
    Performs Binary Search on a sorted list.

    Time Complexity:
    Best Case: O(1)       -> Target is found at the middle element.
    Average Case: O(log n) -> Search space is halved each step.
    Worst Case: O(log n)  -> Element not present or found at the last step.

    Space Complexity:
    O(1) because the algorithm uses constant extra space.

    Parameters:
    arr (list): A sorted list of elements.
    target (int): The value to search for.

    Returns:
    int: Index of the target if found, otherwise -1.
    """

    left = 0
    right = len(arr) - 1

    while left <= right:
        mid = (left + right) // 2

        if arr[mid] == target:
            return mid
        elif arr[mid] < target:
            left = mid + 1
        else:
            right = mid - 1

    return -1
```



```
[6]
✓ 0s
# -------- Test Cases --------
def test_binary_search():
    print("Running Test Cases...\n")

    test_cases = [
        ([1, 3, 5, 7, 9, 11], 7),
        ([2, 4, 6, 8, 10], 2),
        ([10, 20, 30, 40, 50], 35),
        ([], 5),
        ([1], 1),
        ([5, 10, 15, 20, 25], 25)
    ]

    for i, (arr, target) in enumerate(test_cases, 1):
        result = binary_search(arr, target)
        print(f"Test Case {i}:")
        print("Array:", arr)
        print("Target:", target)
        print("Result Index:", result)
        print("-" * 30)


# Run tests
test_binary_search()
```

```
Running Test Cases...

Test Case 1:
Array: [1, 3, 5, 7, 9, 11]
Target: 7
Result Index: 3
------------------------------
Test Case 2:
Array: [2, 4, 6, 8, 10]
Target: 2
Result Index: 0
------------------------------
Test Case 3:
Array: [10, 20, 30, 40, 50]
Target: 35
Result Index: -1
------------------------------
Test Case 4:
Array: []
Target: 5
Result Index: -1
------------------------------
Test Case 5:
Array: [1]
Target: 1
Result Index: 0
------------------------------
Test Case 6:
Array: [5, 10, 15, 20, 25]
Target: 25
Result Index: 4
------------------------------
```

Explaination : The binary_search() function repeatedly checks the middle element of the sorted list and compares it with the target, reducing the search space by half each time until the element is found or the list is exhausted.

Complexity: Best Case O(1), Average & Worst Case O(log n).

Task Description #3: Smart Healthcare Appointment Scheduling System

A healthcare platform maintains appointment records containing appointment ID, patient name, doctor name, appointment time, and consultation fee. The system needs to:

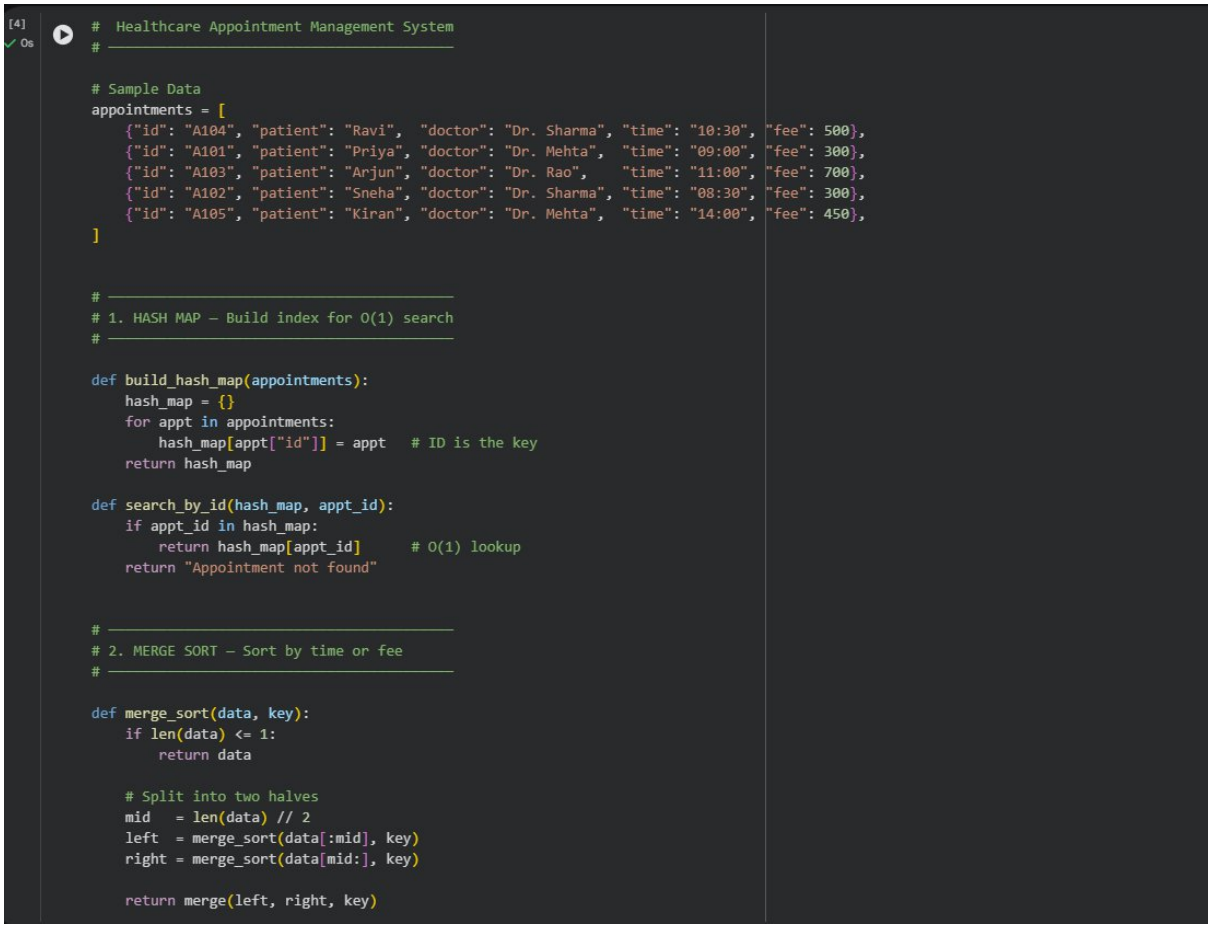
1. Search appointments using appointment ID.
2. Sort appointments based on time or consultation fee.

Student Task

• Use AI to recommend suitable searching and sorting algorithms.

• Justify the selected algorithms.

• Implement the algorithms in Python.



```
[4]
✓ 0s
#  Healthcare Appointment Management System
# ─────────────────────────────────────────

# Sample Data
appointments = [
    {"id": "A104", "patient": "Ravi",  "doctor": "Dr. Sharma", "time": "10:30", "fee": 500},
    {"id": "A101", "patient": "Priya", "doctor": "Dr. Mehta",  "time": "09:00", "fee": 300},
    {"id": "A103", "patient": "Arjun", "doctor": "Dr. Rao",    "time": "11:00", "fee": 700},
    {"id": "A102", "patient": "Sneha", "doctor": "Dr. Sharma", "time": "08:30", "fee": 300},
    {"id": "A105", "patient": "Kiran", "doctor": "Dr. Mehta",  "time": "14:00", "fee": 450},
]


# ─────────────────────────────────────────
# 1. HASH MAP — Build index for O(1) search
# ─────────────────────────────────────────

def build_hash_map(appointments):
    hash_map = {}
    for appt in appointments:
        hash_map[appt["id"]] = appt   # ID is the key
    return hash_map

def search_by_id(hash_map, appt_id):
    if appt_id in hash_map:
        return hash_map[appt_id]      # O(1) lookup
    return "Appointment not found"


# ─────────────────────────────────────────
# 2. MERGE SORT — Sort by time or fee
# ─────────────────────────────────────────

def merge_sort(data, key):
    if len(data) <= 1:
        return data

    # Split into two halves
    mid   = len(data) // 2
    left  = merge_sort(data[:mid], key)
    right = merge_sort(data[mid:], key)

    return merge(left, right, key)
```

[4]
✓ 0s

```
def merge(left, right, key):
    result = []
    i = j = 0

    # Compare and merge
    while i < len(left) and j < len(right):
        if left[i][key] <= right[j][key]:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
            j += 1

    # Add remaining elements
    result.extend(left[i:])
    result.extend(right[j:])
    return result


# ─────────────────────────────────────────
# 3. DISPLAY HELPER
# ─────────────────────────────────────────

def display(appointments):
    print(f"{'ID':<6} {'Patient':<8} {'Doctor':<12} {'Time':<8} {'Fee'}")
    print("-" * 45)
    for a in appointments:
        print(f"{a['id']:<6} {a['patient']:<8} {a['doctor']:<12} {a['time']:<8} ₹{a['fee']}")
    print()


# ─────────────────────────────────────────
# MAIN — Run Everything
# ─────────────────────────────────────────

# Build hash map
hash_map = build_hash_map(appointments)

# Search
print(" Search by Appointment ID: A103")
print(search_by_id(hash_map, "A103"))
print()
```

```python
# Search
print(" Search by Appointment ID: A103")
print(search_by_id(hash_map, "A103"))
print()

print("Search by Appointment ID: A999")
print(search_by_id(hash_map, "A999"))
print()

# Sort by Time
print("Sorted by Appointment Time:")
sorted_by_time = merge_sort(appointments, "time")
display(sorted_by_time)

# Sort by Fee
print("Sorted by Consultation Fee:")
sorted_by_fee = merge_sort(appointments, "fee")
display(sorted_by_fee)
```

```
 Search by Appointment ID: A103
{'id': 'A103', 'patient': 'Arjun', 'doctor': 'Dr. Rao', 'time': '11:00', 'fee': 700}

Search by Appointment ID: A999
Appointment not found

Sorted by Appointment Time:
ID     Patient  Doctor       Time     Fee
-----------------------------------------
A102   Sneha    Dr. Sharma   08:30    ₹300
A101   Priya    Dr. Mehta    09:00    ₹300
A104   Ravi     Dr. Sharma   10:30    ₹500
A103   Arjun    Dr. Rao      11:00    ₹700
A105   Kiran    Dr. Mehta    14:00    ₹450

Sorted by Consultation Fee:
ID     Patient  Doctor       Time     Fee
-----------------------------------------
A101   Priya    Dr. Mehta    09:00    ₹300
A102   Sneha    Dr. Sharma   08:30    ₹300
A105   Kiran    Dr. Mehta    14:00    ₹450
A104   Ravi     Dr. Sharma   10:30    ₹500
A103   Arjun    Dr. Rao      11:00    ₹700
```

Algorithm Selection For Searching → Hash Map (Dictionary)

Appointment IDs are unique → perfect for key-value lookup O(1) average time — instant lookup regardless of data size

For Sorting → Merge Sort

Stable sort → preserves original order for equal values O(n log n) always — consistent performance Works well for both time and fee sorting

Justification : Hash Map is used for searching because it finds any appointment by ID instantly in O(1) time, like a dictionary where you directly jump to the word instead of reading every page. Merge Sort is used for sorting because it always runs in O(n log n) time and is stable, meaning if two patients have the same fee (like ₹300), their original order is preserved — which matters in medical records.

Task Description #4: Railway Ticket Reservation System

Scenario

A railway reservation system stores booking details such as ticket ID, passenger name, train number, seat number, and travel date. The

system must:

1. Search tickets using ticket ID.

2. Sort bookings based on travel date or seat number.

Student Task

• Identify efficient algorithms using AI assistance.

• Justify the algorithm choices.

• Implement searching and sorting in Python.

```
[6]
✓ 0s
    return result

# ─────────────────────────────────────────
# 2. BINARY SEARCH — Search by Ticket ID
# ─────────────────────────────────────────

def binary_search(sorted_data, ticket_id):
    low  = 0
    high = len(sorted_data) - 1

    while low <= high:
        mid = (low + high) // 2
        mid_id = sorted_data[mid]["ticket_id"]

        if mid_id == ticket_id:
            return sorted_data[mid]       # ✅ Found
        elif mid_id < ticket_id:
            low = mid + 1                 # Search right half
        else:
            high = mid - 1               # Search left half

    return "Ticket not found"


# ─────────────────────────────────────────
# 3. DISPLAY HELPER
# ─────────────────────────────────────────

def display(bookings):
    print(f"{'Ticket ID':<12} {'Passenger':<10} {'Train':<8} {'Seat':<6} {'Date'}")
    print("-" * 50)
    for b in bookings:
        print(f"{b['ticket_id']:<12} {b['passenger']:<10} {b['train']:<8} {b['seat']:<6} {b['date']}")
    print()


# ─────────────────────────────────────────
# MAIN — Run Everything
# ─────────────────────────────────────────

# Step 1: Sort by ticket ID first (needed for binary search)
sorted_by_id = merge_sort(bookings, "ticket_id")
```

[4]
✓ 0s

```
def merge(left, right, key):
    result = []
    i = j = 0

    # Compare and merge
    while i < len(left) and j < len(right):
        if left[i][key] <= right[j][key]:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
            j += 1

    # Add remaining elements
    result.extend(left[i:])
    result.extend(right[j:])
    return result


# ─────────────────────────────────────────
# 3. DISPLAY HELPER
# ─────────────────────────────────────────

def display(appointments):
    print(f"{'ID':<6} {'Patient':<8} {'Doctor':<12} {'Time':<8} {'Fee'}")
    print("-" * 45)
    for a in appointments:
        print(f"{a['id']:<6} {a['patient']:<8} {a['doctor']:<12} {a['time']:<8} ₹{a['fee']}")
    print()


# ─────────────────────────────────────────
# MAIN — Run Everything
# ─────────────────────────────────────────

# Build hash map
hash_map = build_hash_map(appointments)

# Search
print(" Search by Appointment ID: A103")
print(search_by_id(hash_map, "A103"))
print()
```

Task Description #5: Smart Hostel Room Allocation System

A hostel management system stores student room allocation details including student ID, room number, floor, and allocation date. The system needs to:

1. Search allocation details using student ID.
2. Sort records based on room number or allocation date.

Student Task

• Use AI to suggest optimized algorithms.

• Justify the selections.

• Implement the solution in Python.

[7]
✓ 0s

```python
# ─────────────────────────────────────────
#  Smart Hostel Room Allocation System
# ─────────────────────────────────────────

# Sample Data
allocations = [
    {"student_id": "S104", "name": "Ravi",  "room": 204, "floor": 2, "date": "2026-01-15"},
    {"student_id": "S101", "name": "Priya", "room": 101, "floor": 1, "date": "2026-01-10"},
    {"student_id": "S103", "name": "Arjun", "room": 301, "floor": 3, "date": "2026-01-20"},
    {"student_id": "S102", "name": "Sneha", "room": 101, "floor": 1, "date": "2026-01-12"},
    {"student_id": "S105", "name": "Kiran", "room": 202, "floor": 2, "date": "2026-01-18"},
]


# ─────────────────────────────────────────
# 1. HASH MAP — Build index for O(1) search
# ─────────────────────────────────────────

def build_hash_map(allocations):
    hash_map = {}
    for record in allocations:
        hash_map[record["student_id"]] = record   # Student ID as key
    return hash_map

def search_by_id(hash_map, student_id):
    if student_id in hash_map:
        return hash_map[student_id]               # O(1) direct lookup
    return "Student record not found"


# ─────────────────────────────────────────
# 2. MERGE SORT — Sort by room or date
# ─────────────────────────────────────────

def merge_sort(data, key):
    if len(data) <= 1:
        return data

    # Split into two halves
    mid   = len(data) // 2
    left  = merge_sort(data[:mid], key)
    right = merge_sort(data[mid:], key)

    return merge(left, right, key)

def merge(left, right, key):
    result = []
    i = j = 0

    # Compare and merge both halves
    while i < len(left) and j < len(right):
        if left[i][key] <= right[j][key]:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
            j += 1

    # Add remaining elements
    result.extend(left[i:])
    result.extend(right[j:])
```

```
# 3. DISPLAY HELPER
# ─────────────────────────────────────

def display(allocations):
    print(f"{'Student ID':<12} {'Name':<8} {'Room':<6} {'Floor':<7} {'Date'}")
    print("-" * 50)
    for a in allocations:
        print(f"{a['student_id']:<12} {a['name']:<8} {a['room']:<6} {a['floor']:<7} {a['date']}")
    print()


# ─────────────────────────────────────
# MAIN — Run Everything
# ─────────────────────────────────────

# Build Hash Map
hash_map = build_hash_map(allocations)

# Search
print("Search Student ID: S103")
print(search_by_id(hash_map, "S103"))
print()

print("Search Student ID: S999")
print(search_by_id(hash_map, "S999"))
print()

# Sort by Room Number
print("Sorted by Room Number:")
sorted_by_room = merge_sort(allocations, "room")
display(sorted_by_room)

# Sort by Allocation Date
print(" Sorted by Allocation Date:")
sorted_by_date = merge_sort(allocations, "date")
display(sorted_by_date)
```

```
Search Student ID: S103
{'student_id': 'S103', 'name': 'Arjun', 'room': 301, 'floor': 3, 'date': '2026-01-20'}

Search Student ID: S999
Student record not found

Sorted by Room Number:
Student ID   Name     Room   Floor   Date
--------------------------------------------------
S101         Priya    101    1       2026-01-10
S102         Sneha    101    1       2026-01-12
S105         Kiran    202    2       2026-01-18
S104         Ravi     204    2       2026-01-15
S103         Arjun    301    3       2026-01-20

 Sorted by Allocation Date:
Student ID   Name     Room   Floor   Date
--------------------------------------------------
S101         Priya    101    1       2026-01-10
S102         Sneha    101    1       2026-01-12
S104         Ravi     204    2       2026-01-15
S105         Kiran    202    2       2026-01-18
S103         Arjun    301    3       2026-01-20
```

Algorithm Selection For Searching → Hash Map (Dictionary)

Student IDs are unique → perfect for direct key-value lookup O(1) average time — instant access regardless of data size

For Sorting → Merge Sort

Stable sort → preserves order for equal room numbers or dates O(n log n) always — consistent and reliable performance

Justification : Hash Map is used for searching because student IDs are unique, so it finds any record in O(1) time instantly — like a locker system where each student has a unique key to their locker. Merge Sort is used for sorting because it always runs in O(n log n) time and is stable, meaning if two students share the same room number (like Priya & Sneha both in Room 101), their original allocation order is safely preserved — which is critical in hostel management.

Task Description #6: Online Movie Streaming Platform

A streaming service maintains movie records with movie ID, title, genre, rating, and release year. The platform needs to:

1. Search movies by movie ID.

2. Sort movies based on rating or release year.

Student Task

• Recommend searching and sorting algorithms using AI.

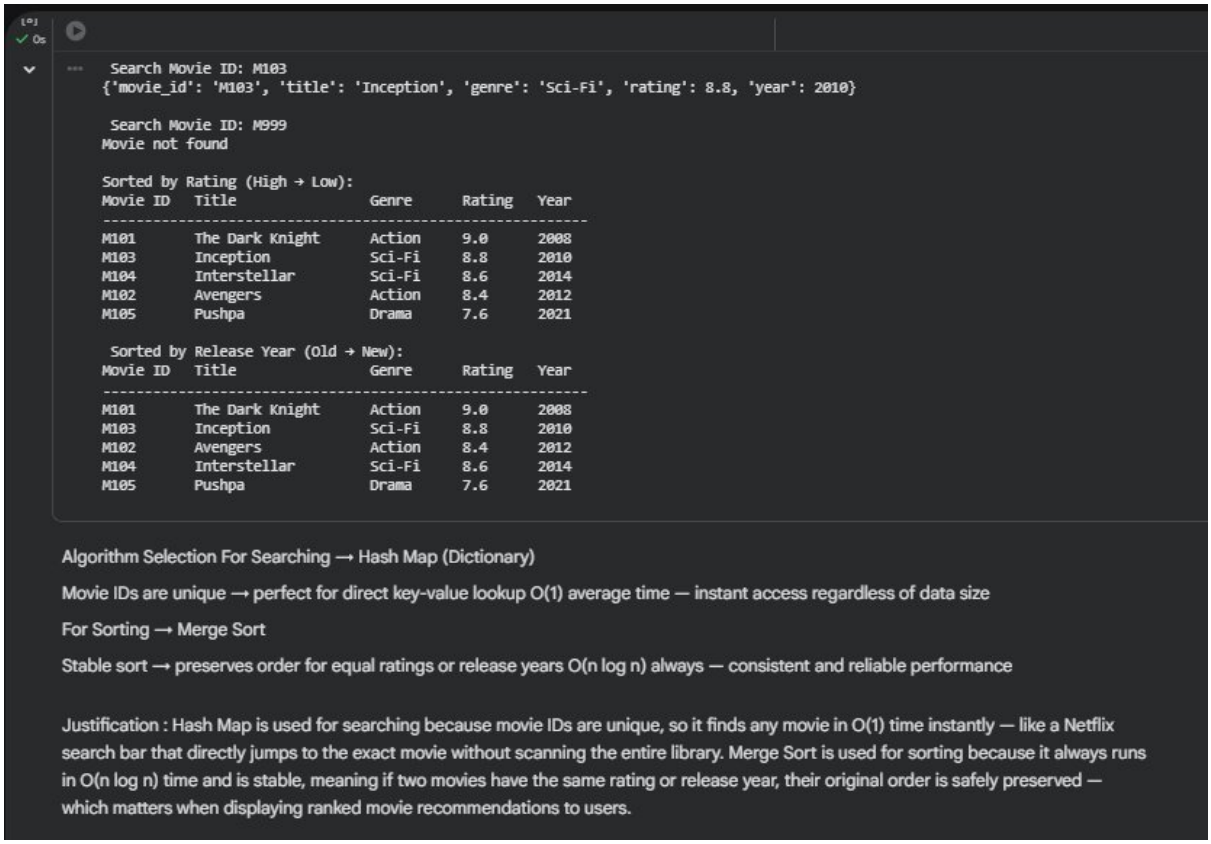
• Justify the chosen algorithms.

• Implement Python functions.

[8] ✓ 0s

```
# ──────────────────────────────────────────
#  Online Movie Streaming Platform
# ──────────────────────────────────────────

# Sample Data
movies = [
    {"movie_id": "M104", "title": "Interstellar",  "genre": "Sci-Fi",   "rating": 8.6, "year": 2014},
    {"movie_id": "M101", "title": "The Dark Knight","genre": "Action",  "rating": 9.0, "year": 2008},
    {"movie_id": "M103", "title": "Inception",      "genre": "Sci-Fi",   "rating": 8.8, "year": 2010},
    {"movie_id": "M102", "title": "Avengers",       "genre": "Action",  "rating": 8.4, "year": 2012},
    {"movie_id": "M105", "title": "Pushpa",         "genre": "Drama",   "rating": 7.6, "year": 2021},
]


# ──────────────────────────────────────────
# 1. HASH MAP — Build index for O(1) search
# ──────────────────────────────────────────

def build_hash_map(movies):
    hash_map = {}
    for movie in movies:
        hash_map[movie["movie_id"]] = movie    # Movie ID as key
    return hash_map

def search_by_id(hash_map, movie_id):
    if movie_id in hash_map:
        return hash_map[movie_id]              # O(1) direct lookup
    return "Movie not found"


# ──────────────────────────────────────────
# 2. MERGE SORT — Sort by rating or year
# ──────────────────────────────────────────

def merge_sort(data, key, reverse=False):
    if len(data) <= 1:
        return data

    # Split into two halves
    mid   = len(data) // 2
    left  = merge_sort(data[:mid], key, reverse)
    right = merge_sort(data[mid:], key, reverse)

    return merge(left, right, key, reverse)

def merge(left, right, key, reverse):
    result = []
    i = j = 0

    # Compare and merge both halves
    while i < len(left) and j < len(right):
        # reverse=True  → descending (highest rating first)
        # reverse=False → ascending  (oldest year first)
        if reverse:
            condition = left[i][key] >= right[j][key]
        else:
            condition = left[i][key] <= right[j][key]

        if condition:
            result.append(left[i])
            i += 1
```

```
        if condition:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
            j += 1

    # Add remaining elements
    result.extend(left[i:])
    result.extend(right[j:])
    return result


# ─────────────────────────────────────────
# 3. DISPLAY HELPER
# ─────────────────────────────────────────

def display(movies):
    print(f"{'Movie ID':<10} {'Title':<20} {'Genre':<10} {'Rating':<8} {'Year'}")
    print("-" * 58)
    for m in movies:
        print(f"{m['movie_id']:<10} {m['title']:<20} {m['genre']:<10} {m['rating']:<8} {m['year']}")
    print()


# ─────────────────────────────────────────
# MAIN — Run Everything
# ─────────────────────────────────────────

# Build Hash Map
hash_map = build_hash_map(movies)

# Search
print(" Search Movie ID: M103")
print(search_by_id(hash_map, "M103"))
print()

print(" Search Movie ID: M999")
print(search_by_id(hash_map, "M999"))
print()

# Sort by Rating (Highest First)
print("Sorted by Rating (High → Low):")
sorted_by_rating = merge_sort(movies, "rating", reverse=True)
display(sorted_by_rating)

# Sort by Release Year (Oldest First)
print(" Sorted by Release Year (Old → New):")
sorted_by_year = merge_sort(movies, "year", reverse=False)
display(sorted_by_year)
```



Task Description #7: Smart Agriculture Crop Monitoring System

An agriculture monitoring system stores crop data with crop ID, crop name, soil moisture level, temperature, and yield estimate. Farmers need to:

1. Search crop details using crop ID.
2. Sort crops based on moisture level or yield estimate.

Student Task

• Use AI-assisted reasoning to select algorithms.

• Justify algorithm suitability.

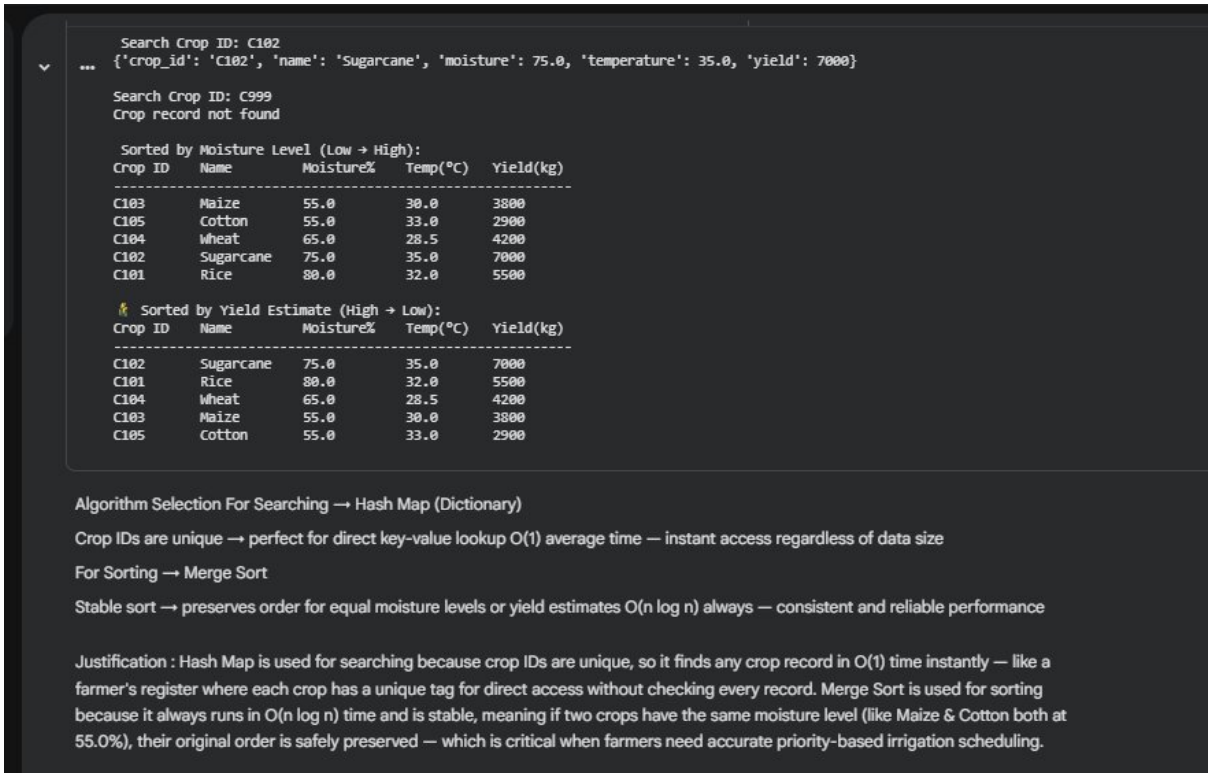
• Implement searching and sorting in Python.

[9] ✓ 0s

```
# ─────────────────────────────────────────
#  Smart Agriculture Crop Monitoring System
# ─────────────────────────────────────────

# Sample Data
crops = [
    {"crop_id": "C104", "name": "Wheat",      "moisture": 65.0, "temperature": 28.5, "yield": 4200},
    {"crop_id": "C101", "name": "Rice",       "moisture": 80.0, "temperature": 32.0, "yield": 5500},
    {"crop_id": "C103", "name": "Maize",      "moisture": 55.0, "temperature": 30.0, "yield": 3800},
    {"crop_id": "C102", "name": "Sugarcane",  "moisture": 75.0, "temperature": 35.0, "yield": 7000},
    {"crop_id": "C105", "name": "Cotton",     "moisture": 55.0, "temperature": 33.0, "yield": 2900},
]


# ─────────────────────────────────────────
# 1. HASH MAP — Build index for O(1) search
# ─────────────────────────────────────────

def build_hash_map(crops):
    hash_map = {}
    for crop in crops:
        hash_map[crop["crop_id"]] = crop       # Crop ID as key
    return hash_map

def search_by_id(hash_map, crop_id):
    if crop_id in hash_map:
        return hash_map[crop_id]               # O(1) direct lookup
    return "Crop record not found"


# ─────────────────────────────────────────
# 2. MERGE SORT — Sort by moisture or yield
# ─────────────────────────────────────────

def merge_sort(data, key, reverse=False):
    if len(data) <= 1:
        return data

    # Split into two halves
    mid   = len(data) // 2
    left  = merge_sort(data[:mid], key, reverse)
    right = merge_sort(data[mid:], key, reverse)

    return merge(left, right, key, reverse)

def merge(left, right, key, reverse):
    result = []
    i = j = 0

    # Compare and merge both halves
    while i < len(left) and j < len(right):
        # reverse=True  → descending (highest yield first)
        # reverse=False → ascending  (lowest moisture first)
        if reverse:
            condition = left[i][key] >= right[j][key]
        else:
            condition = left[i][key] <= right[j][key]

        if condition:
            result.append(left[i])
            i += 1
```

[9]
✓ 0s

```
            else:
                result.append(right[j])
                j += 1

        # Add remaining elements
        result.extend(left[i:])
        result.extend(right[j:])
        return result


# ─────────────────────────────────────────
# 3. DISPLAY HELPER
# ─────────────────────────────────────────

def display(crops):
    print(f"{'Crop ID':<10} {'Name':<12} {'Moisture%':<12} {'Temp(°C)':<10} {'Yield(kg)'}")
    print("-" * 58)
    for c in crops:
        print(f"{c['crop_id']:<10} {c['name']:<12} {c['moisture']:<12} {c['temperature']:<10} {c['yield']}")
    print()


# ─────────────────────────────────────────
# MAIN — Run Everything
# ─────────────────────────────────────────

# Build Hash Map
hash_map = build_hash_map(crops)

# Search
print(" Search Crop ID: C102")
print(search_by_id(hash_map, "C102"))
print()

print("Search Crop ID: C999")
print(search_by_id(hash_map, "C999"))
print()

# Sort by Moisture Level (Low → High)
print(" Sorted by Moisture Level (Low → High):")
sorted_by_moisture = merge_sort(crops, "moisture", reverse=False)
display(sorted_by_moisture)

# Sort by Yield Estimate (High → Low)
print(" 🌾 Sorted by Yield Estimate (High → Low):")
sorted_by_yield = merge_sort(crops, "yield", reverse=True)
display(sorted_by_yield)
```



Task Description #8: Airport Flight Management System

An airport system stores flight information including flight ID, airline name, departure time, arrival time, and status. The system must:

1. Search flight details using flight ID.
2. Sort flights based on departure time or arrival time.

Student Task

• Use AI to recommend algorithms.

• Justify the algorithm selection.

• Implement searching and sorting logic in Python.

[10] ✓ 0s

```
# ──────────────────────────────────────
#  Airport Flight Management System
# ──────────────────────────────────────

# Sample Data
flights = [
    {"flight_id": "F104", "airline": "IndiGo",     "departure": "10:30", "arrival": "13:00", "status": "On Time"},
    {"flight_id": "F101", "airline": "Air India",  "departure": "08:00", "arrival": "11:30", "status": "Delayed"},
    {"flight_id": "F103", "airline": "SpiceJet",   "departure": "09:45", "arrival": "12:15", "status": "On Time"},
    {"flight_id": "F102", "airline": "Vistara",    "departure": "08:00", "arrival": "10:30", "status": "Cancelled"},
    {"flight_id": "F105", "airline": "GoFirst",    "departure": "14:00", "arrival": "17:30", "status": "On Time"},
]


# ──────────────────────────────────────
# 1. HASH MAP — Build index for O(1) search
# ──────────────────────────────────────

def build_hash_map(flights):
    hash_map = {}
    for flight in flights:
        hash_map[flight["flight_id"]] = flight    # Flight ID as key
    return hash_map

def search_by_id(hash_map, flight_id):
    if flight_id in hash_map:
        return hash_map[flight_id]                # O(1) direct lookup
    return "Flight not found"


# ──────────────────────────────────────
# 2. MERGE SORT — Sort by departure or arrival
# ──────────────────────────────────────

def merge_sort(data, key, reverse=False):
    if len(data) <= 1:
        return data

    # Split into two halves
    mid   = len(data) // 2
    left  = merge_sort(data[:mid], key, reverse)
    right = merge_sort(data[mid:], key, reverse)

    return merge(left, right, key, reverse)

def merge(left, right, key, reverse):
    result = []
    i = j = 0

    # Compare and merge both halves
    while i < len(left) and j < len(right):
        # reverse=True  → descending (latest time first)
        # reverse=False → ascending  (earliest time first)
        if reverse:
            condition = left[i][key] >= right[j][key]
        else:
            condition = left[i][key] <= right[j][key]

        if condition:
            result.append(left[i])
            i += 1
```

```
        if condition:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
            j += 1

    # Add remaining elements
    result.extend(left[i:])
    result.extend(right[j:])
    return result


# ─────────────────────────────────────────
# 3. DISPLAY HELPER
# ─────────────────────────────────────────

def display(flights):
    print(f"{'Flight ID':<11} {'Airline':<12} {'Departure':<11} {'Arrival':<10} {'Status'}")
    print("-" * 58)
    for f in flights:
        print(f"{f['flight_id']:<11} {f['airline']:<12} {f['departure']:<11} {f['arrival']:<10} {f['status']}")
    print()


# ─────────────────────────────────────────
# MAIN — Run Everything
# ─────────────────────────────────────────

# Build Hash Map
hash_map = build_hash_map(flights)

# Search
print(" Search Flight ID: F103")
print(search_by_id(hash_map, "F103"))
print()

print(" Search Flight ID: F999")
print(search_by_id(hash_map, "F999"))
print()

# Sort by Departure Time (Earliest First)
print(" Sorted by Departure Time (Early → Late):")
sorted_by_departure = merge_sort(flights, "departure", reverse=False)
display(sorted_by_departure)

# Sort by Arrival Time (Earliest First)
print("Sorted by Arrival Time (Early → Late):")
sorted_by_arrival = merge_sort(flights, "arrival", reverse=False)
display(sorted_by_arrival)
```

```
display(sorted_by_arrival)
```

```
 Search Flight ID: F103
{'flight_id': 'F103', 'airline': 'SpiceJet', 'departure': '09:45', 'arrival': '12:15', 'status': 'On Time'}

 Search Flight ID: F999
Flight not found

 Sorted by Departure Time (Early → Late):
Flight ID   Airline      Departure   Arrival    Status
------------------------------------------------------------
F101        Air India    08:00       11:30      Delayed
F102        Vistara      08:00       10:30      Cancelled
F103        SpiceJet     09:45       12:15      On Time
F104        IndiGo       10:30       13:00      On Time
F105        GoFirst      14:00       17:30      On Time

Sorted by Arrival Time (Early → Late):
Flight ID   Airline      Departure   Arrival    Status
------------------------------------------------------------
F102        Vistara      08:00       10:30      Cancelled
F101        Air India    08:00       11:30      Delayed
F103        SpiceJet     09:45       12:15      On Time
F104        IndiGo       10:30       13:00      On Time
F105        GoFirst      14:00       17:30      On Time
```

Algorithm Selection For Searching → Hash Map (Dictionary)

Flight IDs are unique → perfect for direct key-value lookup O(1) average time — instant access regardless of data size

For Sorting → Merge Sort

Stable sort → preserves order for equal departure or arrival times O(n log n) always — consistent and reliable performance

Justification : Hash Map is used for searching because flight IDs are unique, so it finds any flight in O(1) time instantly — like an airport display board where entering a flight number directly shows its gate and status without scanning every flight. Merge Sort is used for sorting because it always runs in O(n log n) time and is stable, meaning if two flights have the same departure time (like Air India & Vistara both at 08:00), their original order is safely preserved — which is critical in airports to avoid scheduling conflicts and runway mismanagement.